

Software Requirements Specifications

Project Title: Micro-Service Hub

Part 4:Non-Functional Requirements

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4. Non-Functional Requirements

4.1 Security

- User passwords and personal information must be stored securely.
- Unauthorized access to accounts and sensitive data must be prevented.
- NID and payment information must remain confidential.
- Proper authentication and access control must be implemented.

4.2 Performance

- The system should respond quickly to user requests.
- Search, service posting, and application processes should have minimal delay.
- The system should support multiple users simultaneously without major performance issues.

4.3 Usability

- The interface should be simple, clear, and user-friendly.
- Users should easily post, search, and apply for services.
- The platform should be responsive on desktop, tablet, and mobile devices.

4.4 Reliability

- User, service, application, and payment data must be stored accurately.
- The system should minimize data loss and maintain consistent service status.
- Transactions and service records should be reliable.

4.5 Availability

- The platform should be accessible whenever required.
- System downtime should be minimized.
- The system should handle temporary service or API failures appropriately.

4.6 Scalability

- The system should support increasing numbers of users and services.

- New service categories and features should be added without major architectural changes.

4.7 Maintainability

- The system should use modular and well-organized code.
- Bugs should be easy to identify and fix.
- Future updates and feature additions should be easy to implement.

4.8 Compatibility

- The platform should work on major modern browsers such as Chrome, Firefox, Edge.
- It should support desktop, laptop, tablet, and mobile devices.

4.9 Data Integrity

- The system must maintain accurate and consistent data.
- Invalid and duplicate data should be prevented.
- Service, application, and payment records must remain synchronized.

4.10 Privacy

- Personal, NID, and communication information must be protected.
- Sensitive information should only be accessible to authorized users.

4.11 Extensibility

- The system should allow future features such as mobile apps, AI-based matching, digital wallets, and additional payment methods.

4.12 Legal and Ethical Requirements

- User data must be handled responsibly.
- The system should support reporting of fraud, harassment, and other inappropriate activities.
- User safety and privacy should be maintained.